A.3.2 Switch

Many programming languages also offer a specialized instruction to be used when the condition we check can assume several discrete values; usually this instruction is called `switch`, and allows to enumerate possible values that an expression can assume and perform a different `case` branch for each one of them (plus a `default` branch if the value doesn't match any of the other cases).

```
switch x² + y²
  case 0
    label ← "origin"
  case 1
    label ← "unit circle"
  default
    label ← "circle"
```

The code above evaluates the expression $x^2 + y^2$: each case statement corresponds to a different branch executed when the expression evaluates to whatever follows the keyword, so if $x^2 + y^2$ evaluates to `0`, then the code after the first `case` is executed, and the variable label is set to "`origin`"; similarly, if the value is `1`, label is set to "`unit circle`". For any other value, the branch after the keyword default is executed and label is set to "`circle`".

We will use an extended version of this instruction, allowing also ranges to be specified in case statements: